Returns Process — Team Manual (Business + System)

*Combined handover guide. Covers what each team **does** (business process) and **how the systems support it** (Garage, Redline, PWA scanner). Standalone for now; to be folded into the Garage and Redline page-manuals once the process is implemented. Owner: Store (intake) + Production (repair). Last updated: 2026-06-06.*

1. The big picture (read this first)

- **The Store is the single door for ALL returns.** Every returned unit — modern, legacy, damaged, switcheroo, loose car or remote — enters through Store intake first. Nothing skips it.
- **Returns arrive in shipments.** One shipment can carry any mix of products, channels, and conditions.
- **The Store sorts every unit into one of four piles** (Store's own judgement — there is no enforced rule):
- **UDR** — Undamaged Return → re-dispatch as-is, **no repair**.
- **CXR** — Customer Return → **repair**.
- **BRV** — Bulk/Vendor Return → **repair**.
- **Loss** — switcheroo / wrong item / nothing received → **written off** (record what arrived).
- **Cars and remotes are independent units in returns.** During returns, a car and a remote are treated separately. Any old car↔remote link is **broken at Store intake** (kept as history, never erased) and a **fresh pairing is made at the repair QC-PASS station**, exactly like a brand-new production run.
- **Five golden rules** (never break these):

1. Single door — Store intake is the only entry.
2. Nothing leaves the Store without a **box label**.
3. A raw **legacy EAN never travels downstream** — it gets a real LOT box label at intake.
4. A **UDR box is never opened** — preserve it, re-dispatch on its existing label.
5. Car↔remote pairing is re-made **only at QC PASS**; the old link is preserved as history.

2. Who uses what

Team	System	Does
Store	**Garage** (returns console) + **PWA scanner** (Return Intake station)	Receive shipments, scan + classify every unit into UDR/CXR/BRV/Loss, relabel legacy units, issue units out
Production	**Redline** (see piles, request repair runs) + **PWA scanner** (repair stations)	Request stock from Store, run the repair line, push finished units to dispatch
Dispatch	(unchanged)	Receives UDRs and finished repaired units via the normal handover

3. STORE — receiving & sorting (Garage + PWA scanner)

1. **Open a return shipment** in Garage → it gets an `RS-NNN` number. Enter **courier** and **date received**. Garage shows a **shipment barcode** on screen.

2. **Bind the scanner**: on the PWA, pick **Return Intake**, and **scan the shipment barcode** once. The device is now feeding *this* shipment. (Two shipments can be processed at two stations at once — each scans its own barcode.)

3. **Scan each unit in**, in this priority:

- Box intact + LOT box label readable → **scan the box label**.
- Box damaged / box label unreadable → **scan the car (or remote) UPC**.
- No readable LOT at all → **scan the product EAN/barcode**.

- Nothing scannable → **enter the product manually** in Garage. Each scanned unit pops up as a row in the Garage shipment (a few seconds after the beep — the screen refreshes itself; you don't reload).

4. **Set the disposition** on each row in Garage: **UDR / CXR / BRV / Loss**.

- **Legacy unit** (no LOT label): mark it **Legacy** and **print a fresh box label** from Garage. For a sealed undamaged box the label is applied to the *closed* box (no opening). The printed UPC is automatically **blocked from future batch printing** so no one reprints it later. If the box is opened, attach fresh labels to the **car and/or remote** as received (they stay separate — see §6).
- **Switcheroo / wrong item / nothing received** → **Loss**, and **record what physically arrived** in the note.

5. Each unit also captures: **channel, product, variant, colour, date processed**.
6. **Close the shipment** when the pile is done. Units now sit in their piles, visible to Production in Redline.

> *The shipment **stays open** the whole time you're scanning and dispositioning — it's a live draft. You can close the laptop and resume later; reopening `RS-NNN` shows everything scanned so far.*

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4. STORE — issuing units to Production (scan-out)

Production requests stock; the Store issues it. Issuing **scans each unit out** (just like a normal store pick-list issue). The pick list is **whole units** — some carry box labels, some original car+remote UPCs, some newly-attached legacy UPCs.

- **Issue as Repair** → feeds a repair run (`REP-NNN`). For **CXR + BRV** units.
- **Issue as UDR** → no repair run; each UDR is scanned out and goes straight to PKG OUT.

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5. PRODUCTION — the two paths

5a. UDR path (no repair)

- **One scan at PKG OUT** (the box label) → **handover to dispatch**. The box is **never opened**. It rides its existing (or Store-printed legacy) box label out. That's the whole path.

5b. CXR / BRV path (repair run, `REP-NNN`)

Cars and remotes move through as **independent units** (no pairing yet):

1. **Inspection / Rep Start** — scan the unit, choose **Repair** or **Scrap**.

- *Scrap* → goes to the **scrap pile**, stops here.
- *Repair* → continues.

2. **Repair** — fix the unit, then send to QC (no extra scan).
3. **QC** — same as a normal production run:

- **Pass** → **PKG IN** → **PKG OUT** → handover to dispatch. *(Car↔remote pairing is made here at QC PASS, exactly like a fresh production run.)*
- **Fail** → **WKS IN** → **WKS OUT** → **back to QC**, and repeat until it passes.

4. **Mismatched car/remote counts are normal.** If the Store sends 100 cars but only 70 remotes, Production builds/requests the other 30 remotes from the Store ad-hoc and pairs them at QC PASS.

> *Planned later (not in v1): a unit that fails QC 3 times will auto-scrap. For now the QC loop is unlimited, same as normal production.*

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6. Why cars and remotes are separate in returns

A returned unit may arrive car-only, remote-only, or as a pair whose halves are in different condition. So returns treat the **car and the remote as independent items**. Any existing car↔remote link is **broken when the Store takes the unit in** (the history of the old pairing is kept, never deleted). A **new pairing is created at the repair QC-PASS station** — the same moment a fresh production run pairs a car with its remote. UDRs (sealed, never opened) keep their original pairing.

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7. REDLINE — Production's view

- **Read-only view of the piles** (UDR / CXR / BRV counts by product / variant / colour) + scrap.
- **Request a repair run** → the Store issues the units against it.

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8. Quick reference — disposition → destination

Pile	Condition	Destination
UDR	Undamaged, box sealed	Issue as UDR → 1 scan at PKG OUT → dispatch
CXR	Customer return, needs work	Repair run → Inspect → Repair → QC → PKG → dispatch
BRV	Bulk/vendor return, needs work	Repair run (same as CXR)
Loss	Switcheroo / wrong / not received	Written off; record what arrived
Scrap	Unrepairable (decided at Inspection or QC)	Scrap pile